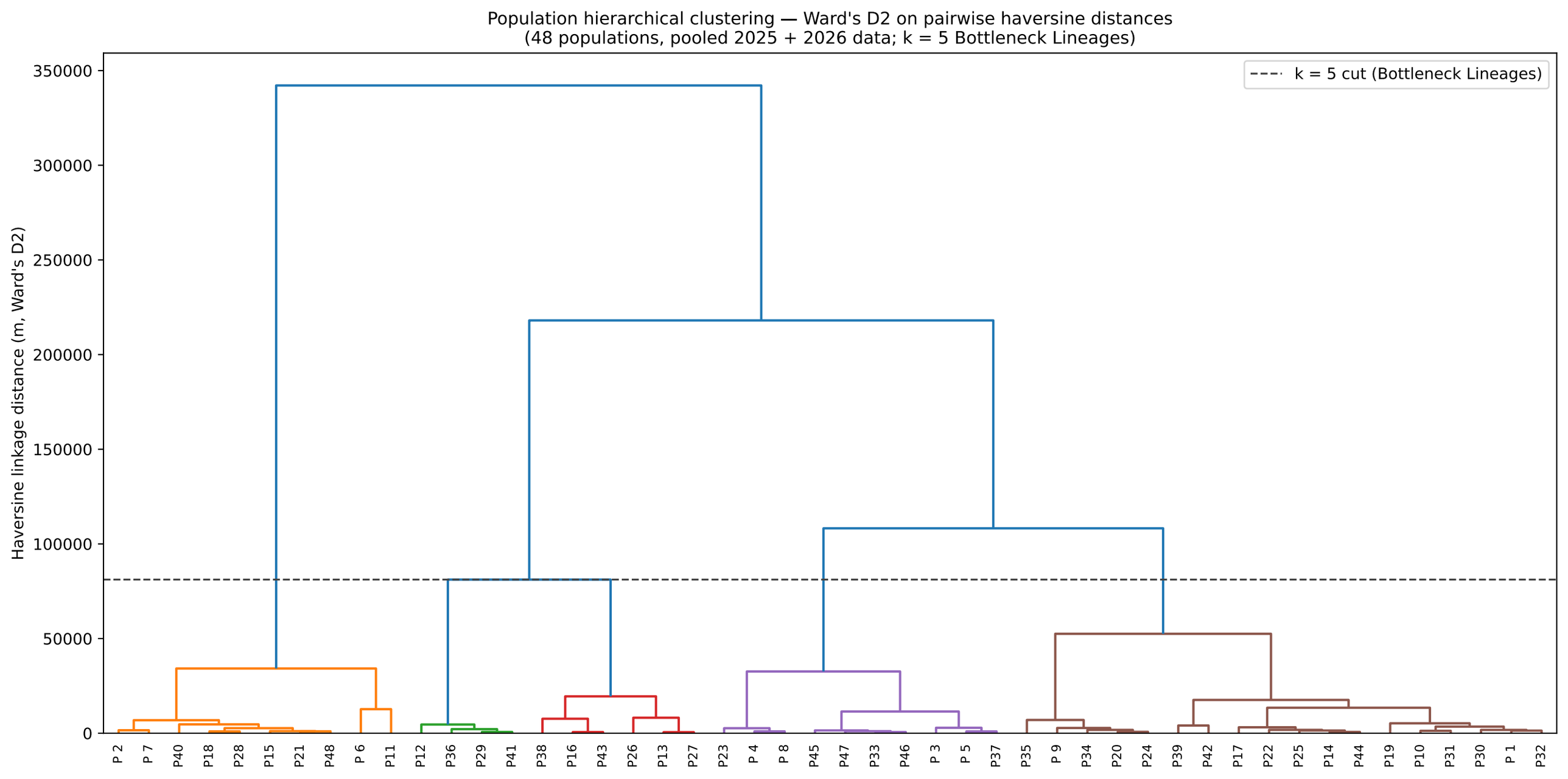
Population hierarchical clustering — Ward's D2 on pairwise haversine distances
(48 populations, pooled 2025 + 2026 data; k = 5 Bottleneck Lineages)
k = 5 cut (Bottleneck Lineages)
Haversine linkage distance (m, Ward's D2)
350000
300000
250000
200000
150000
100000
50000
0
P 2
P 7
P40
P18
P28
P15
P21
P48
P 6
P11
P12
P36
P29
P41
P38
P16
P43
P26
P13
P27
P23
P 4
P 8
P45
P47
P33
P46
P 3
P 5
P37
P35
P 9
P34
P20
P24
P39
P42
P17
P22
P25
P14
P44
P19
P10
P31
P30
P 1
P32